

Portfolio item 3: Report Skeleton

Riley Webster

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1 Introduction

- Define transients in astronomy, focusing on SNe Ia, explaining why they are important as they are standardisable candles.
- Outline the motivations for improving the standardisation capabilities and then develop a clearer understanding of explosion mechanics.
- Mention how current standardisation methods leave a scatter in the residuals that appears to be systematic, thus can be corrected for, resulting in cleaner data for cosmological studies. Specifically, the limits of the Tripp equation and how it doesn't reduce the significance of multiple correlations with the supernova environments. By determining and analysing the possible progenitor channels, not only can more be understood about the physics of the explosion, but this improved understanding can lead to more precise standardisation.
- Introduce LSST, Rubin and Fink as the tools that will provide the means of data collection, specifically the immense volume of alerts providing much better chances of finding early excess flux SNe Ia, and how the Fink broker will provide the processing means to manage the data and enable the data to be used for analysis.

2 Supernova Ia physics and Progenitor Channels

2.1 Type Ia Supernovae

- Describe what type Ia SNe are, explain that they are thermonuclear explosions of carbon-oxygen (C-O) white dwarfs, and how they are standardisable candles, which leads into the standardisation section. Keep this section brief, refer to Ruiter and Seitenzahl (2024)

2.2 Standardisation

- Explaining the use of type Ia supernova as standardisable candles to make distance measurements that are useful in determining cosmological parameters such as the equation of state parameter for dark energy and via determinations of H_0, Ω_m, and Ω_Λ. Introduce the mass-step discovered by Sullivan et al. (2010) and then discuss the studies that have improved standardisation using correlations with more local properties such as lsSFR(Rigault et al., 2020), [OII] emission (Martin et al., 2024),the connection to cosmological parameters (Brout et al., 2022) and how corrections for stretch in SNe Ia need to be accounted for based on environment leading to a non linear correction (Ginolin et al., 2025).

2.3 Progenitor Channels

- Describe the two popular progenitor channels for type Ia supernovae, the single and double degenerate scenarios, and how they differ in the light curve data through modelling with different nickel 56 masses and distributions. Keep brief,

- Primary references are Deckers et al. (2022) and Ruiter and Seitenzahl (2024)

2.4 Early Flux Excess Supernovae

- Here will be an explanation of how important early observation, well before peak flux/magnitude of the supernova, is crucial in having light curves that can help determine the progenitor channel.

- Describe the early excess flux observed in a few of the SNe Ia and how they are characterised, such as through visual inspection and calculation of light curve residuals in mag and flux space.

- Discuss some of the proposed mechanisms that could explain the observed early excess in the light curves, such as ^{56}Ni masses and distributions, interactions with the SNe ejecta and the companion star, and interactions with the circumstellar material.

- Primary references will be Burke et al. (2022), Deckers et al. (2022), and Lim et al. (2023)

3 Host Galaxy Environment and SN Ia light curves

3.1 Host Galaxy Properties

- Discuss topics like what is known about where SNe Ia occur most radially in the Host galaxy, and what bands SNe Ia trace in galaxy light such as the isophotal analysis done by Pritchet et al. (2024).

- Discuss the host galaxy stellar mass step correction and proposed physical explanations for the observed correlation, as an extrinsic property like host galaxy mass is not going to directly affect the brightness of the SNe Ia.

- Describe the observed differences in correlation between global properties, such as specific star formation rate and the local specific star formation rate, and lower masses and higher SFR in the local environment correlates with SNe Ia properties, such as colour, and how analysis of local environments shows a homogeneity in SNe Ia properties.

- Primary references will include Ginolin et al. (2025), Kelsey et al. (2022), and Ramaiya et al. (2025)

3.2 The Environment and Early Excess Flux

- If early excess flux SNe Ia are found in specific environments, not only will that assist in determining potential progenitor channels. It will also assist in the cutting of SNe Ia in samples used in cosmology, where the early excess SNe Ia could contaminate the cosmology sample.

- The lack of a statistically significant sample size has made it so that early excess flux SNe Ia rates haven't been put as a function of host galaxy properties and thus have eluded systematic classification.

- Primary references will include Burke et al. (2022) and Deckers et al. (2022)

4 Fink, Rubin and LSST

- Give an overview of the specifications of LSST and Rubin telescope, including the survey strategy used for the LSST survey, and describe the volume of alerts that are expected per night. Leading to a discussion on why this volume of alerts requires a broker to process data for science.

- Describe the Fink broker and give an overview of how it helps deliver on the science it'll be used for, namely the classification pipeline of alerts. Including topics like Fink's active learning approach to classifiers and the broker's performance.

- Compare the ZTF survey data that Fink has been using to the data LSST is producing, and how through the higher cadence and greater depth of LSST will detect SNe Ia earlier thus creating statistically significant samples of EEx SNe Ia.

- Primary references for this section will be Fraga et al. (2024), Leoni et al. (2022), and Möller et al. (2021)

5 Project Aims

- State the primary aim of the project, which is to use LSST data collected via the Rubin telescope to identify and classify early excess SNe Ia and then investigate how their incidence correlates with host galaxy properties.

- Attempt to use the early excess flux SNe Ia to make constraints on possible progenitor channels.

6 Planned Methodology

- Use Fink, it searches through alerts that have been classified as SN Ia candidates.

- Follow the methodology of Deckers et al. (2022) to identify the early excess flux SNe Ia, specifically around the approximate 5 days pre-maximum flux and distinguishing between red and blue SNe Ia events. This will be done by using the SALT3 light curve fitting model for better wavelength representation in the UV and NIR to determine residuals with observed fluxes to identify and classify EEx SNe Ia.

- To assign SNe Ia with a host, we will use the Directional Light Radius method, which takes the separation from the SN to the nearest galaxy and then normalises it by the galaxy's directional light radius, which is the radius of a line drawn from a galaxy's centre to a nearby transient event such as a SNe Ia. Gupta et al., 2016

- Finally, compare early excess supernovae Ia rates to host galaxies' properties such as sSFR, galaxy stellar mass, and colour. We will also see how the colour of the SNe Ia varies across the sample to determine if any connections crop up with the host properties, with the colour of early excess SNe Ia. This will be used to make conclusions about whether early excess SNe Ia prefer different host galaxy environments and how that affects the possible progenitor channels for the early excess SNe Ia.

7 References

- This will be were the references are!

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